



J.J. College of Engineering and Technology

Department of Mechanical Engineering

ME25CO8

Metallurgy and Materials Science

Student Notes

Unit - II: Heat Treatment

Scheduled Hours: 9 Periods

Academic Year : 2026–2027

Year and Semester : 2nd Yr and III Sem

Course Handling Faculty

Dr. Sreearravind M.

Assistant Professor

Department of Mechanical Engineering

J.J. College of Engineering and Technology

Contents

Course Objectives	iv
Full Course Content	iv
Course Outcomes	v
Unit II Learning Focus	v
Engineering Context, Indian Relevance and Innovation Link	vi
Text Books	vii
References	vii
1 Introduction to Unit II	1
2 Basic Principles of Heat Treatment	1
2.1 Definition	1
2.2 Basic Steps	1
2.3 Purpose of Heat Treatment	2
2.4 Important Critical Temperatures in Steel	2
3 Annealing Processes	3
3.1 Annealing - General Introduction	3
3.2 Full Annealing	4
3.3 Stress Relief Annealing	5
3.4 Recrystallisation Annealing	5
3.5 Spheroidising Annealing	6
3.6 Comparison of Annealing Processes	7
4 Normalising, Hardening and Tempering of Steel	7
4.1 Normalising	7
4.2 Hardening	8
4.3 Tempering	9
4.4 Comparison of Normalising, Hardening and Tempering	9
5 Isothermal Transformation and CCT Diagrams	10

5.1	Need for Transformation Diagrams	10
5.2	Isothermal Transformation Diagram	10
5.3	Cooling Curves Superimposed on I.T. Diagram	11
5.4	Continuous Cooling Transformation Diagram	11
5.5	CCT Diagram Superimposed on I.T. Diagram	12
5.6	Difference Between I.T. and CCT Diagrams	13
6	Austempering and Martempering	14
6.1	Austempering	14
6.2	Martempering	15
6.3	Comparison of Austempering and Martempering	16
7	Hardenability and Jominy End Quench Test	16
7.1	Hardenability	16
7.2	Hardness and Hardenability	17
7.3	Factors Affecting Hardenability	17
7.4	Jominy End Quench Test	18
7.5	Jominy Test Procedure	18
7.6	Interpretation of Jominy Curve	19
8	Case Hardening and Diffusion Treatments	20
8.1	Case Hardening	20
8.2	Carburizing	21
8.3	Nitriding	22
8.4	Cyaniding	23
8.5	Carbonitriding	23
8.6	Comparison of Case Hardening Methods	24
9	Flame, Induction, Vacuum and Plasma Hardening	25
9.1	Flame Hardening	25
9.2	Induction Hardening	25
9.3	Difference Between Flame Hardening and Induction Hardening	26
9.4	Vacuum Hardening	26
9.5	Plasma Hardening	27
9.6	Vacuum Hardening and Plasma Hardening - Comparison	28
10	Thermo-Mechanical Treatment	28

11 Elementary Ideas on Sintering	29
11.1 Basic Steps in Powder Metallurgy and Sintering	29
11.2 What Happens During Sintering?	29
12 Important Exam-Oriented Points	30
12.1 Very Short Answer Points	30
12.2 Important Two-Mark Questions	31
12.3 Important Five-Mark Questions	32
12.4 Important Long-Answer Questions	32
13 Unit Summary	33
14 Glossary of Important Terminologies	33

Course Objectives

1. To learn the construction of phase diagrams and the use of the Iron–Iron carbide phase diagram for understanding microstructure formation.
2. To learn the selection and application of various heat treatment processes and their influence on microstructure formation.
3. To illustrate the different types of ferrous and non-ferrous alloys and their uses in the engineering field.
4. To illustrate different polymers, ceramics and composites and their uses in the engineering field.
5. To learn various testing procedures and failure mechanisms in engineering materials.

Full Course Content

Course Code: ME25CO8

Course Title: Metallurgy and Materials Science

Unit	Content
UNIT I	CONSTITUTION OF ALLOYS AND PHASE DIAGRAMS Constitution of alloys – Solid solutions, substitutional and interstitial phase diagrams, isomorphous, eutectic, eutectoid, peritectic, and peritectoid reactions, Iron–Iron carbide equilibrium diagram. Classification of steel and cast-iron microstructure, properties and application.
UNIT II	HEAT TREATMENT Definition – Full annealing, stress relief, recrystallisation and spheroidising – normalizing, hardening and tempering of steel. Isothermal transformation diagrams – cooling curves superimposed on I.T. diagram – continuous cooling transformation (CCT) diagram – Austempering, Martempering – Hardenability, Jominy end quench test – case hardening, carburizing, nitriding, cyaniding, carbonitriding – Flame and Induction hardening – Vacuum and Plasma hardening – Thermo-mechanical treatments – elementary ideas on sintering.
UNIT III	FERROUS AND NON-FERROUS METALS Effect of alloying additions on steel (Mn, Si, Cr, Mo, Ni, V, Ti and W) – stainless and tool steels – HSLA – Maraging steels – grey, white, malleable, spheroidal alloy cast irons, copper and its alloys – brass, bronze and cupronickel – aluminium and its alloys; Al–Cu precipitation strengthening treatment – titanium alloys, Mg-alloys, Ni-based super alloys – shape memory alloys – properties and applications – overview of materials standards.
UNIT IV	NON-METALLIC MATERIALS

Unit	Content
UNIT V	Polymers – types of polymers, commodity and engineering polymers – properties and applications of PE, PP, PS, PVC, PMMA, PET, PC, PA, ABS, PAI, PPO, PPS, PEEK, PTFE, thermoset polymers, urea and phenol formaldehydes – nylon, engineering ceramics – properties and applications of Al_2O_3, SiC, Si_3N_4, PSZ and SIALON – intermetallics – composites – matrix and reinforcement materials – applications of composites – nano composites. MECHANICAL PROPERTIES AND DEFORMATION MECHANISMS Mechanisms of plastic deformation, slip and twinning – types of fracture – fracture mechanics – Griffith's theory – testing of materials under tension, compression and shear loads – hardness tests (Brinell, Vickers and Rockwell), micro and nano-hardness tests, impact test Izod and Charpy, fatigue and creep failure mechanisms.

Total: 45 periods (9 periods each)

Course Outcomes

At the end of the course, the students would be able to:

1. Explain alloys and phase diagrams, the Iron–Iron carbon diagram and steel classification.
2. Explain isothermal transformation, continuous cooling diagrams and different heat treatment processes.
3. Clarify the effect of alloying elements on ferrous and non-ferrous metals.
4. Summarize the properties and applications of non-metallic materials.
5. Explain the testing of mechanical properties.

Unit II Learning Focus

For examination preparation, Unit II should be studied with the following sequence:

1. Basic principle of heat treatment: heating, soaking and cooling.
2. Annealing group: full annealing, stress relief annealing, recrystallisation annealing and spheroidising.
3. Normalising, hardening and tempering: purpose, process cycle, final structure and applications.
4. Transformation diagrams: I.T. / TTT diagram, cooling curves and CCT diagram.
5. Special heat treatments: austempering, martempering, hardenability and Jominy end quench test.

6. Surface and case hardening methods: carburizing, nitriding, cyaniding, carbonitriding, flame and induction hardening.
7. Modern and related processes: vacuum hardening, plasma hardening, thermo-mechanical treatment and sintering.

Engineering Context, Indian Relevance and Innovation Link

Why this unit matters: Heat treatment is one of the most important industrial operations for steels. A component may have the correct chemical composition and correct shape, but it may still fail if the microstructure is not properly controlled by heat treatment.

Indian manufacturing relevance

- Heat-treated steels are widely used in automotive components, railway parts, agricultural machinery, construction equipment, pumps, valves, machine tools and defence-related mechanical systems.
- Gears, shafts, springs, dies, crankshafts, camshafts, bearings and cutting tools require controlled hardness, strength, toughness and wear resistance.
- Case hardening and surface hardening are especially important where components need a hard surface and a tough core, such as gears, cams and shafts.
- Sintering is linked with powder metallurgy and is useful for manufacturing small, complex and near-net-shape components with reduced material wastage.

Start-up and project ideas linked with Unit II

1. **Heat-treatment selector:** A simple software tool that suggests annealing, normalising, hardening, tempering or case hardening based on component requirement.
2. **Jominy curve digital record system:** A small application to store Rockwell hardness values at different distances from the quenched end and automatically plot hardenability curves.
3. **Workshop heat-treatment checklist:** A QR-code based record system for heating temperature, soaking time, quenching medium and tempering condition.
4. **Surface hardening decision aid:** A tool to compare carburizing, nitriding, flame hardening and induction hardening for gears, shafts and pins.
5. **Powder metallurgy learning kit:** A teaching kit showing powder mixing, compaction, green compact and sintered product using simple demonstration materials.

High-technology facts related to Unit II

- Modern automotive steels use controlled transformation of austenite to ferrite, pearlite, bainite, martensite and retained austenite.
- Induction hardening is important for localized surface hardening in high-volume manufacturing.
- Vacuum heat treatment is used for precision dies, moulds, aerospace parts and tools where surface oxidation must be avoided.

- Thermo-mechanical treatment combines deformation and heat treatment to obtain fine grains and high strength.
- Data-driven heat treatment and digital manufacturing use temperature-time records, cooling curves, hardness maps and microstructure data.

Text Books

1. Kenneth G. Budinski and Michael K. Budinski, *Engineering Materials*, Prentice Hall of India Private Limited, 9th Edition, 2018.
2. Sydney H. Avner, *Introduction to Physical Metallurgy*, McGraw Hill Book Company, 1994.

References

1. A. Alavudeen, N. Venkateshwaran, and J. T. Winowlin Jappes, *A Textbook of Engineering Materials and Metallurgy*, Laxmi Publications, 2006.
2. Amandeep Singh Wadhwa and Harvinder Singh Dhaliwal, *A Textbook of Engineering Material and Metallurgy*, University Sciences Press, 2008.
3. G. S. Upadhyay and Anish Upadhyay, *Materials Science and Engineering*, Viva Books Pvt. Ltd., New Delhi, 2020.
4. V. Raghavan, *Materials Science and Engineering*, Prentice Hall of India Pvt. Ltd., 6th Edition, 2019.
5. William D. Callister, *Materials Science and Engineering*, Wiley India Pvt. Ltd., 2nd Edition Reprint, 2019.

1 Introduction to Unit II

Heat treatment is the controlled process of heating and cooling a metal or alloy in the solid state to obtain required changes in microstructure and mechanical properties. In steels, heat treatment is mainly used to control the transformation of austenite into products such as pearlite, bainite and martensite.

The same steel can behave differently after different heat treatments. For example, a steel component may be softened for machining by full annealing, strengthened by normalising, hardened by quenching, toughened by tempering, or surface hardened by carburizing or induction hardening.

Exam point: In heat treatment answers, always connect process → microstructure → properties → applications.

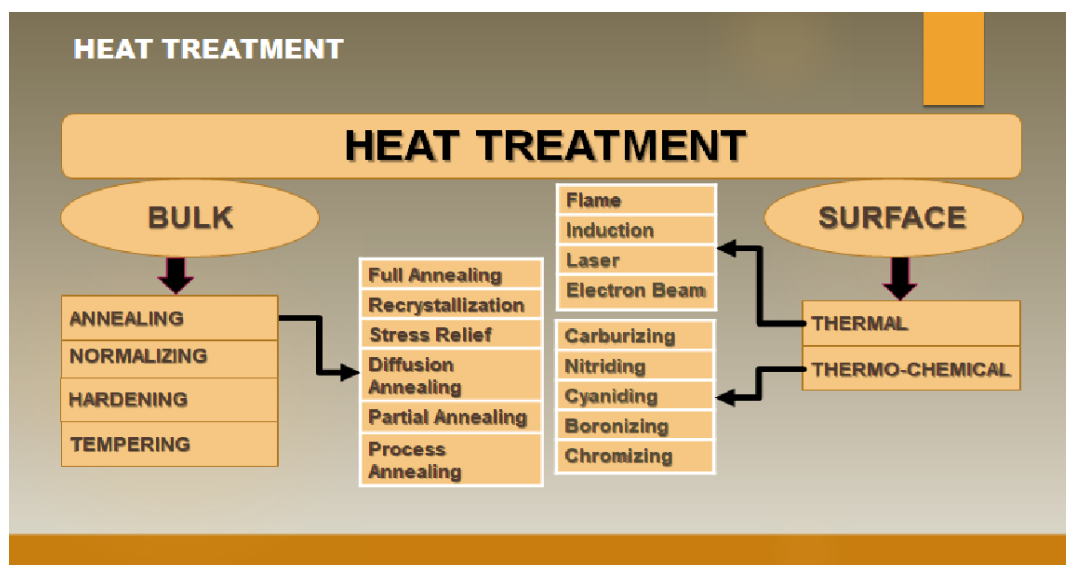


Figure 1: Overall relationship between heat treatment, microstructure, properties and applications.

2 Basic Principles of Heat Treatment

2.1 Definition

Heat treatment is a controlled process of heating and cooling a metal or alloy in the solid state to change its microstructure and improve properties such as hardness, strength, ductility, toughness, machinability, wear resistance and dimensional stability.

2.2 Basic Steps

Most heat treatment processes involve three basic stages:

1. **Heating:** The component is heated to the required temperature.
2. **Soaking:** The component is held at that temperature for uniform heating and required

transformation.

3. **Cooling:** The component is cooled at a controlled rate using furnace cooling, air cooling, oil quenching, water quenching or gas quenching.

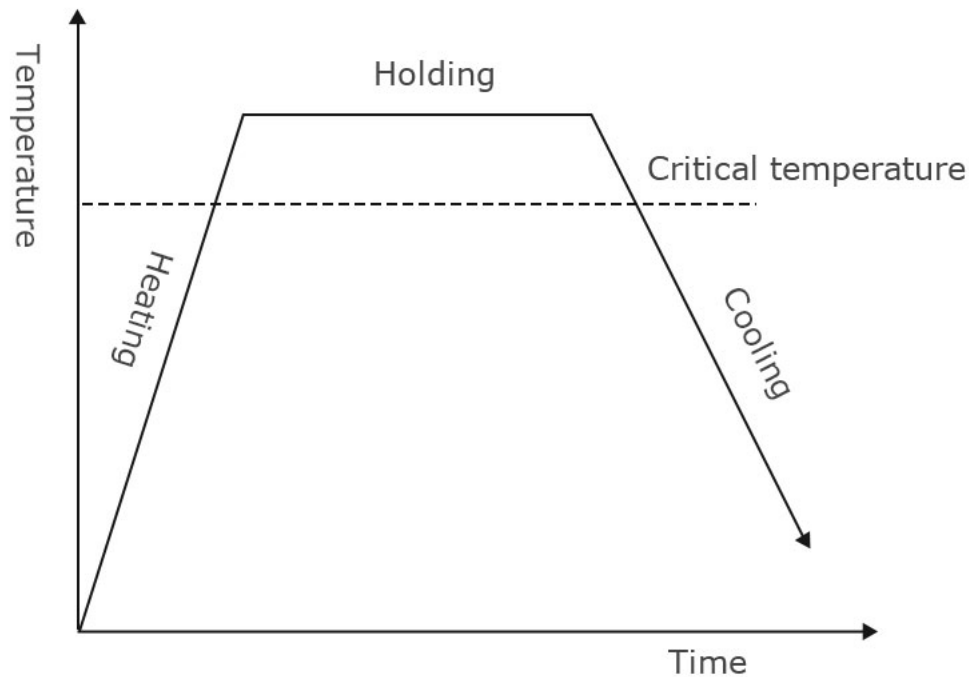


Figure 2: Basic heat treatment cycle showing heating, soaking and cooling.

2.3 Purpose of Heat Treatment

Heat treatment is carried out to:

1. Improve hardness and strength.
2. Improve ductility and toughness.
3. Relieve internal stresses.
4. Improve machinability and formability.
5. Refine grain structure.
6. Improve wear resistance and fatigue resistance.
7. Obtain desired microstructures such as coarse pearlite, fine pearlite, bainite, martensite and tempered martensite.

2.4 Important Critical Temperatures in Steel

For steel heat treatment, the Iron–Iron carbide diagram is used as a guide for choosing heating ranges.

Symbol	Meaning
A_1	Lower critical temperature or eutectoid temperature, approximately 727°C.
A_3	Upper critical temperature for hypo-eutectoid steels.
A_{cm}	Upper critical temperature boundary for hyper-eutectoid steels.
M_s	Martensite start temperature.
M_f	Martensite finish temperature.

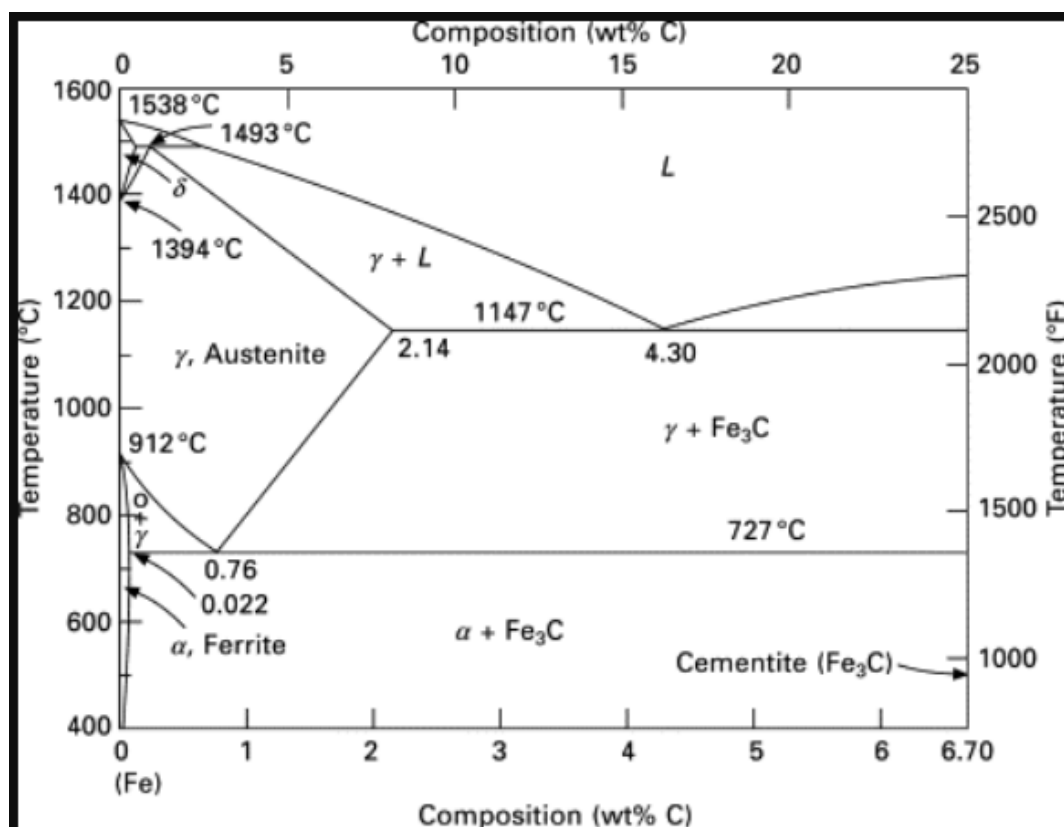


Figure 3: Heat treatment temperature ranges shown on the steel portion of the Iron–Iron carbide diagram.

3 Annealing Processes

3.1 Annealing - General Introduction

Annealing is a heat treatment process in which a metal or alloy is heated to a suitable temperature, held at that temperature for sufficient time and cooled slowly, usually inside the furnace.

Main purposes of annealing are:

- to soften the material,
- to improve ductility,
- to improve machinability,

- to relieve internal stresses,
- to refine or modify microstructure,
- to reduce hardness after cold working or hardening.

Exam point: The key word for annealing is **slow cooling**.

3.2 Full Annealing

Full annealing is a heat treatment process in which steel is heated above the critical temperature, held for sufficient time and cooled very slowly in the furnace to obtain a soft, ductile and refined microstructure.

Heating range:

- Hypo-eutectoid steel: 30–50°C above A_3 .
- Hyper-eutectoid steel: 30–50°C above A_1 , generally below A_{cm} .

Final microstructure:

- Hypo-eutectoid steel: ferrite + coarse pearlite.
- Eutectoid steel: coarse pearlite.
- Hyper-eutectoid steel: pearlite + cementite.

Applications: castings, forgings, rolled steel products and steel parts before machining.

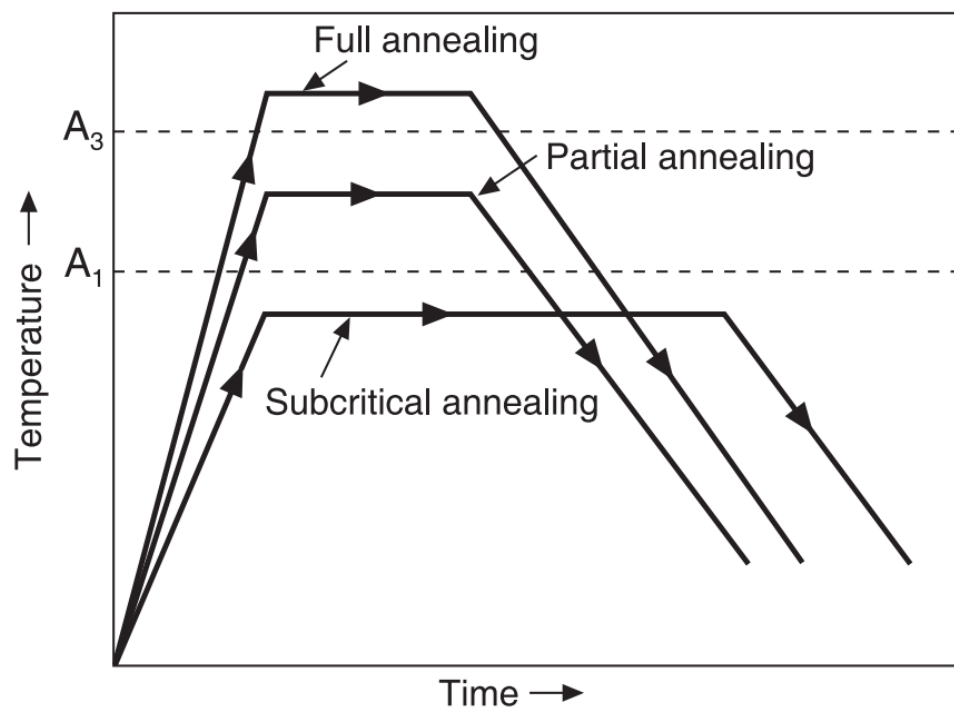


Figure 4: Full annealing cycle showing heating, soaking and furnace cooling.

3.3 Stress Relief Annealing

Stress relief annealing is a process in which steel is heated below A_1 , held for sufficient time and cooled slowly to remove internal residual stresses without major change in microstructure.

Residual stresses may be produced by: welding, casting, forging, rolling, machining, quenching and uneven cooling.

Important points:

- It is usually done below 727°C for steels.
- There is no major phase transformation.
- Hardness and microstructure remain almost unchanged.
- Distortion and cracking tendency are reduced.

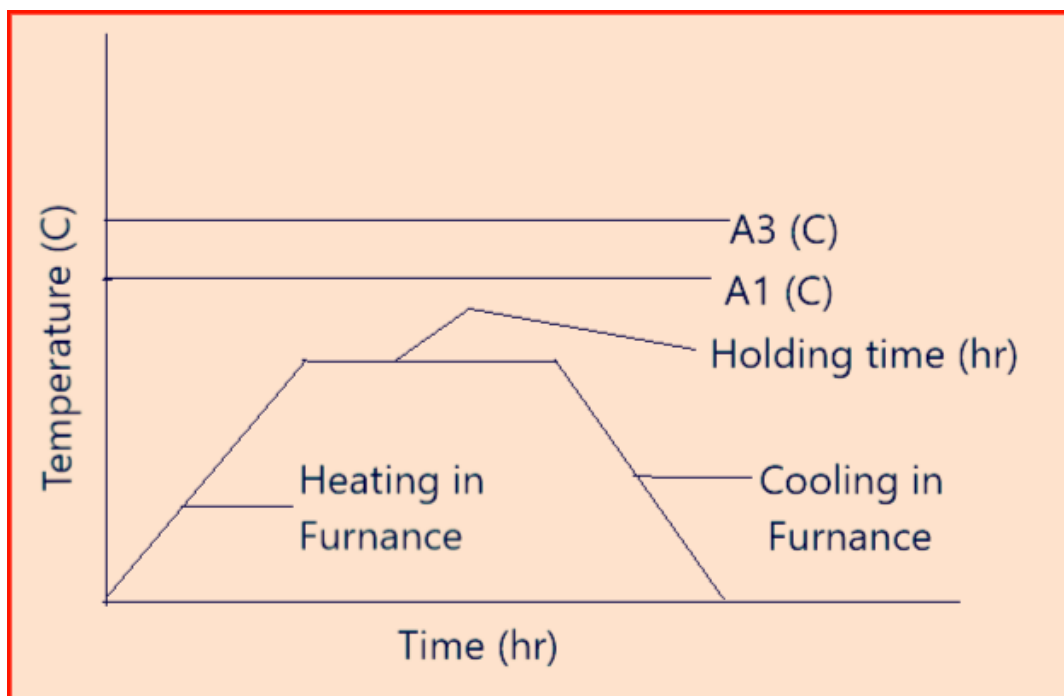


Figure 5: Stress relief annealing cycle below the lower critical temperature.

3.4 Recrystallisation Annealing

Recrystallisation annealing is a heat treatment process in which a cold-worked metal is heated to a temperature at which new strain-free grains form and replace the deformed grains.

Purpose:

1. Remove work hardening.
2. Restore ductility.
3. Reduce hardness.
4. Improve formability.
5. Form new equiaxed and strain-free grains.

Used after: cold rolling, wire drawing, sheet forming, cold forging and deep drawing.

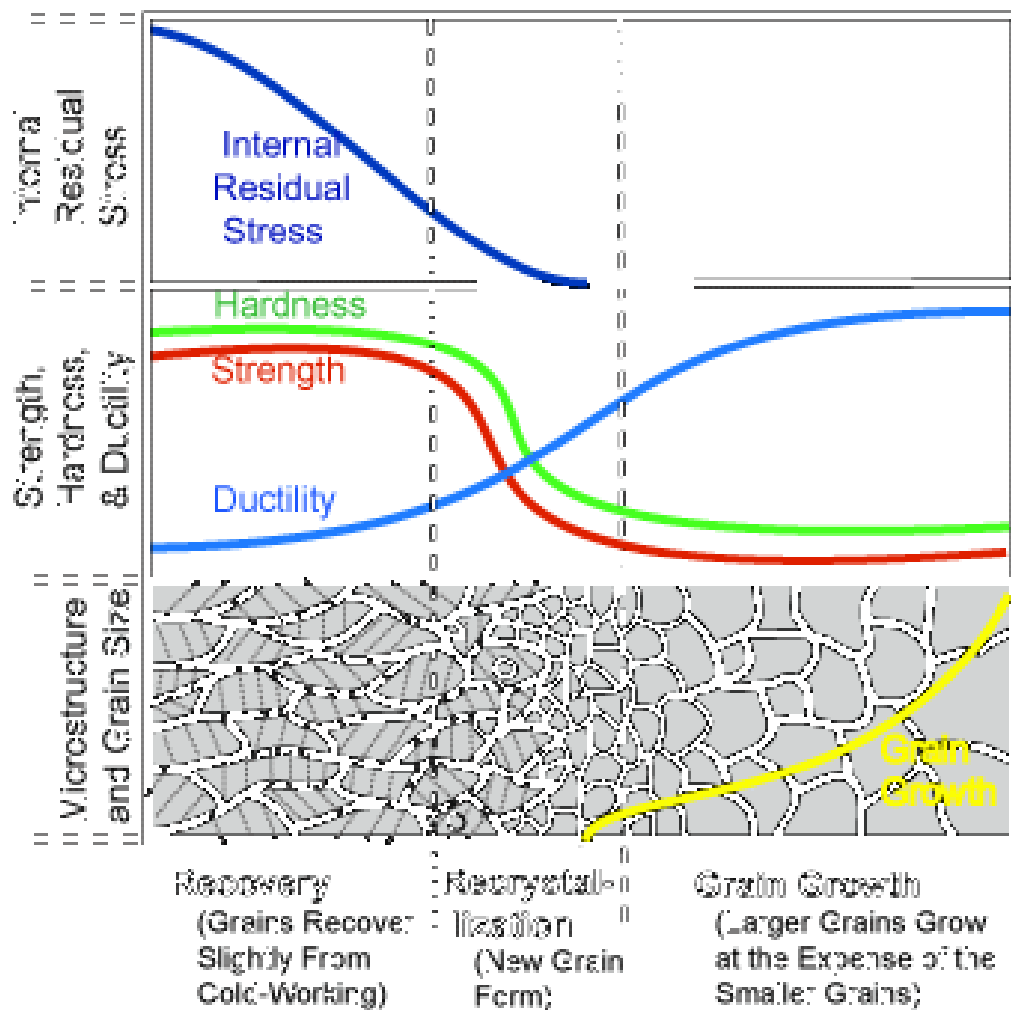


Figure 6: Change from elongated cold-worked grains to new strain-free recrystallised grains.

3.5 Spheroidising Annealing

Spheroidising is an annealing process in which high-carbon steel is heated and held near A_1 for a long time to convert lamellar cementite into small spherical cementite particles in a ferrite matrix.

Purpose:

- Improve machinability of high-carbon steels.
- Reduce hardness.
- Improve ductility.
- Prepare tool steels and bearing steels for machining before hardening.

Final structure: spheroidite, which consists of spherical cementite particles in ferrite matrix.

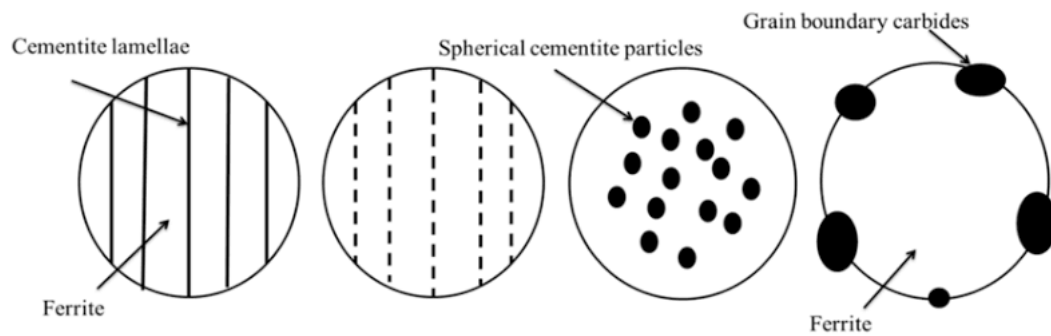


Figure 7: Spheroidising: conversion of lamellar cementite into spheroidal cementite particles.

3.6 Comparison of Annealing Processes

Process	Heating range	Main purpose	Main result
Full annealing	Above critical temperature	Softening and machinability	Coarse pearlite, high ductility
Stress relief annealing	Below A_1	Remove residual stress	Stress reduction with little structural change
Recrystallisation annealing	Above recrystallisation temperature	Remove cold work effect	New strain-free grains
Spheroidising	Near A_1 for long time	Machinability of high-carbon steel	Spherical cementite in ferrite

4 Normalising, Hardening and Tempering of Steel

4.1 Normalising

Normalising is a heat treatment process in which steel is heated above the critical temperature, held for sufficient time and cooled in still air.

Purpose:

- Refine grain size.
- Improve strength and toughness.
- Produce uniform microstructure.
- Remove the effects of previous working.

Compared with full annealing, normalising gives a finer pearlite structure because air cooling is faster than furnace cooling. Therefore, normalised steel usually has higher strength and hardness than fully annealed steel.

Difference between Annealing and normalising

Annealing	Normalising
1. Metal is heated and is left at the same furnace for cooling.	Metal is heated and held at for same time and then allowed to cool down freely in the air.
2. Annealing is done before machining.	Normalising is done after forging or wrong heat treatment.
3. Tensile strength is reduced after annealing.	Tensile strength is increased after normalizing.
4. Grain size is changed to larger in the internal structure.	Grain size is changed to very small in the internal structure.

Figure 8: Comparison of furnace cooling in annealing and air cooling in normalising.

4.2 Hardening

Hardening is a heat treatment process in which steel is heated to the austenitic region, held for sufficient time and rapidly cooled by quenching to obtain high hardness.

Common quenching media: water, brine, oil, polymer solution and forced air for selected steels.

Final structure: martensite.

Important point: Martensite is very hard but brittle. Therefore, hardened steel is usually tempered before use.

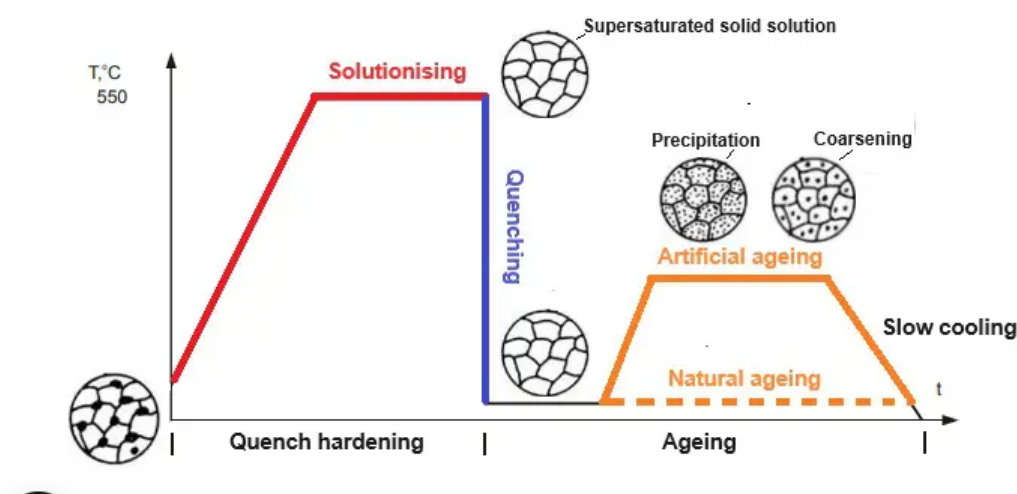


Figure 9: Hardening cycle showing austenitising followed by rapid quenching.

4.3 Tempering

Tempering is a heat treatment process in which hardened steel is reheated below A_1 , held for some time and cooled to reduce brittleness and improve toughness.

Purpose:

1. Reduce brittleness of martensite.
2. Improve toughness and ductility.
3. Reduce internal stresses.
4. Retain useful hardness.
5. Produce tempered martensite.

Exam point: Tempering is normally done after hardening. Higher tempering temperature generally decreases hardness and increases toughness.

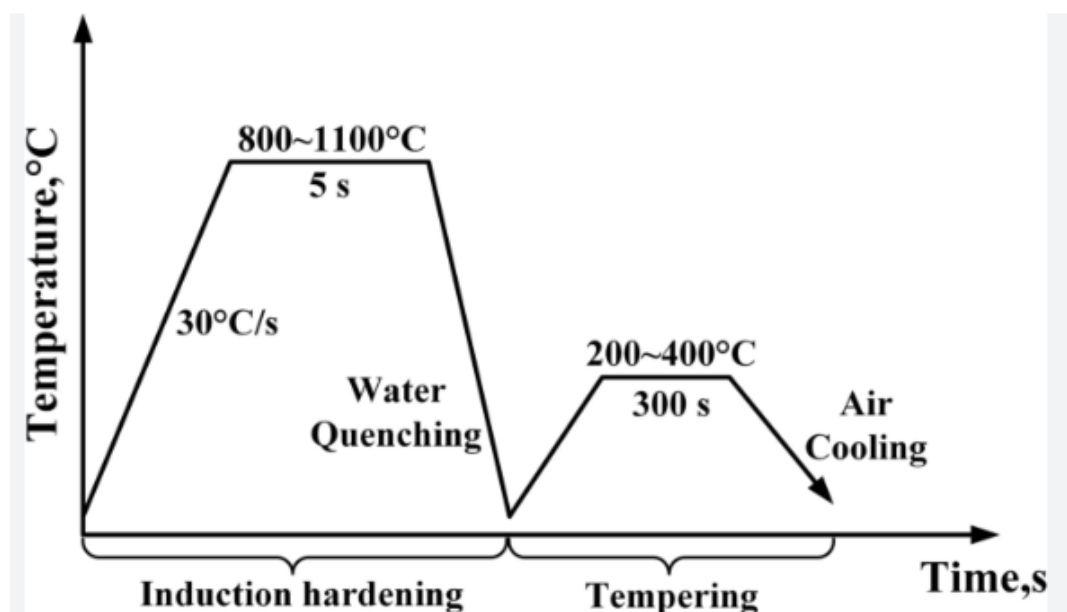


Figure 10: Sequence of hardening and tempering showing martensite and tempered martensite.

4.4 Comparison of Normalising, Hardening and Tempering

Process	Cooling method	Main product	Purpose
Normalising	Air cooling	Fine pearlite with ferrite/cementite depending on carbon content	Grain refinement, strength and toughness
Hardening	Rapid quenching	Martensite	High hardness and wear resistance
Tempering	Reheating below A_1 and cooling	Tempered martensite	Reduce brittleness and improve toughness

5 Isothermal Transformation and CCT Diagrams

5.1 Need for Transformation Diagrams

The Iron–Iron carbide diagram gives equilibrium phase information. However, heat treatment depends strongly on time and cooling rate. Therefore, transformation diagrams are required to understand how austenite transforms with respect to time and temperature.

5.2 Isothermal Transformation Diagram

An isothermal transformation diagram shows the transformation of austenite at constant temperature with respect to time. It is also called I.T. diagram or TTT diagram.

Main parts of I.T. / TTT diagram:

- transformation start curve,
- transformation finish curve,
- nose of the curve,
- pearlite region,
- bainite region,
- M_s and M_f lines.

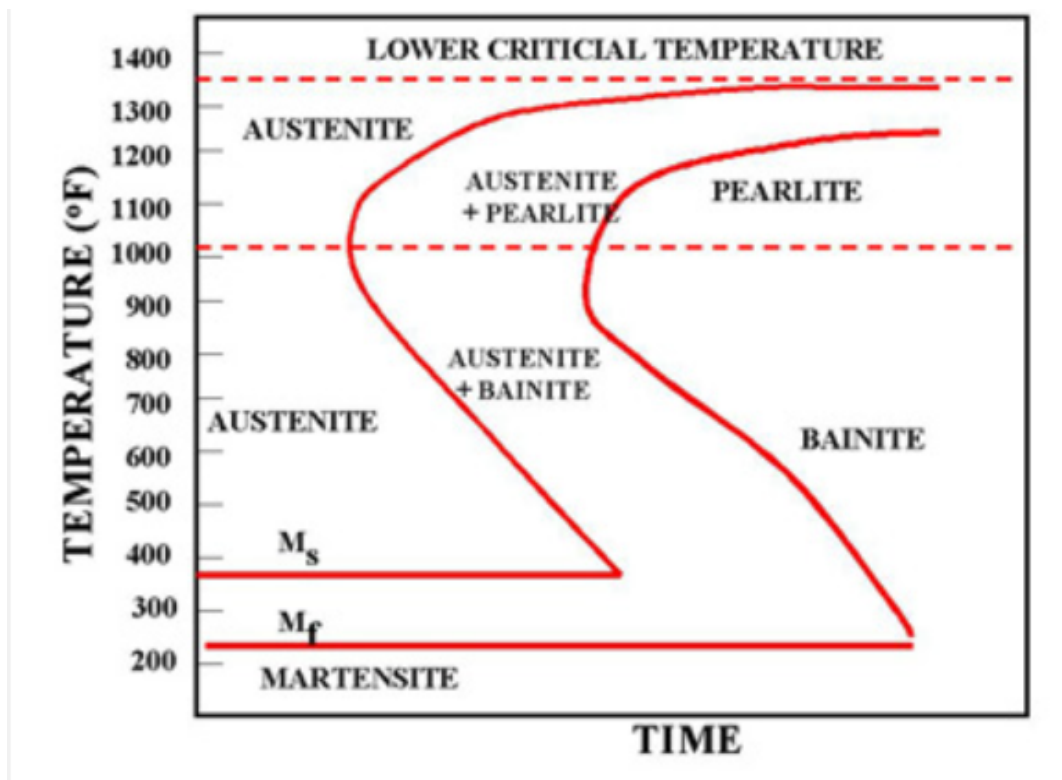


Figure 11: Basic I.T. / TTT diagram showing pearlite, bainite and martensite regions.

5.3 Cooling Curves Superimposed on I.T. Diagram

A cooling curve shows the decrease of temperature with time during cooling. When cooling curves are drawn on an I.T. diagram, the final microstructure can be predicted.

Cooling condition	Transformation product	Heat treatment link
Very slow cooling	Coarse pearlite	Full annealing
Moderate air cooling	Fine pearlite	Normalising
Faster cooling	Bainite + martensite possible	Special cooling conditions
Very rapid cooling avoiding the nose	Martensite	Hardening

Critical cooling rate is the minimum cooling rate required to avoid the nose of the TTT diagram and produce martensite.

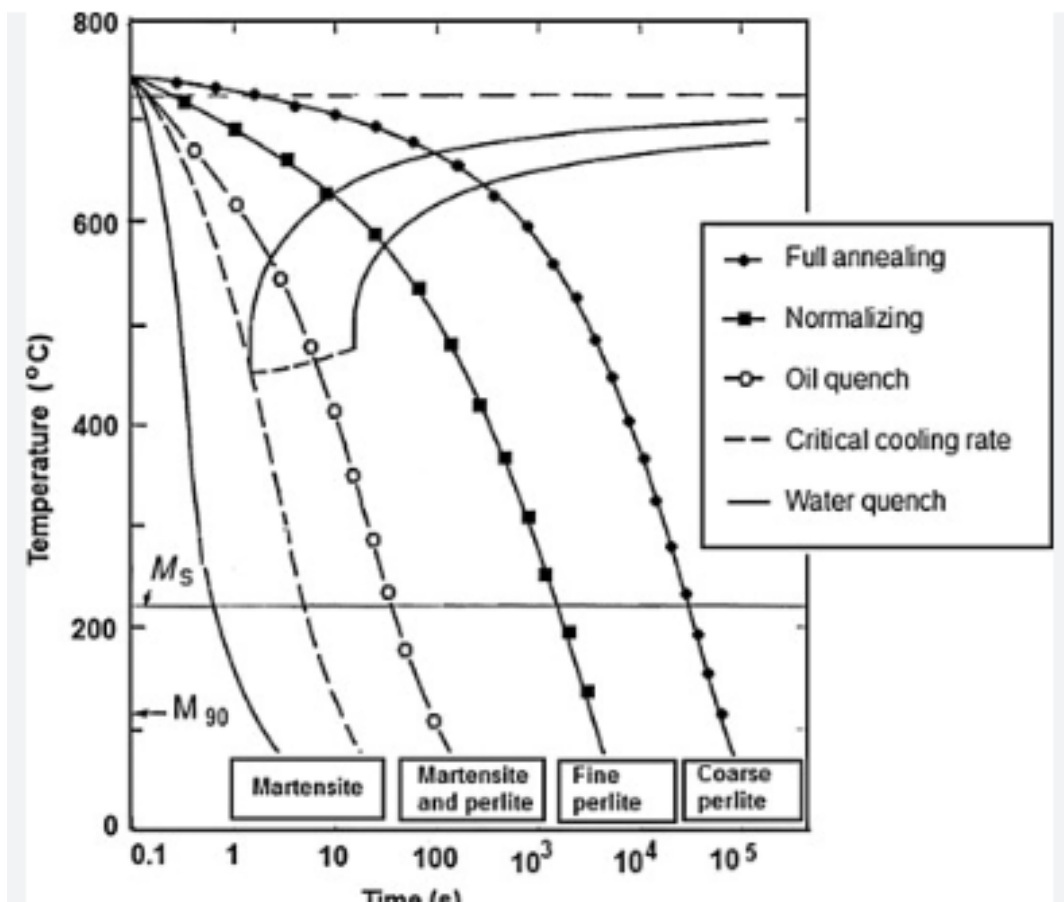


Figure 12: Cooling curves superimposed on I.T. / TTT diagram showing different transformation products.

5.4 Continuous Cooling Transformation Diagram

A Continuous Cooling Transformation diagram shows the transformation of austenite during continuous cooling. It is called the CCT diagram.

CCT diagrams are more practical than I.T. diagrams because most industrial heat treatment operations involve continuous cooling rather than holding at a constant temperature.

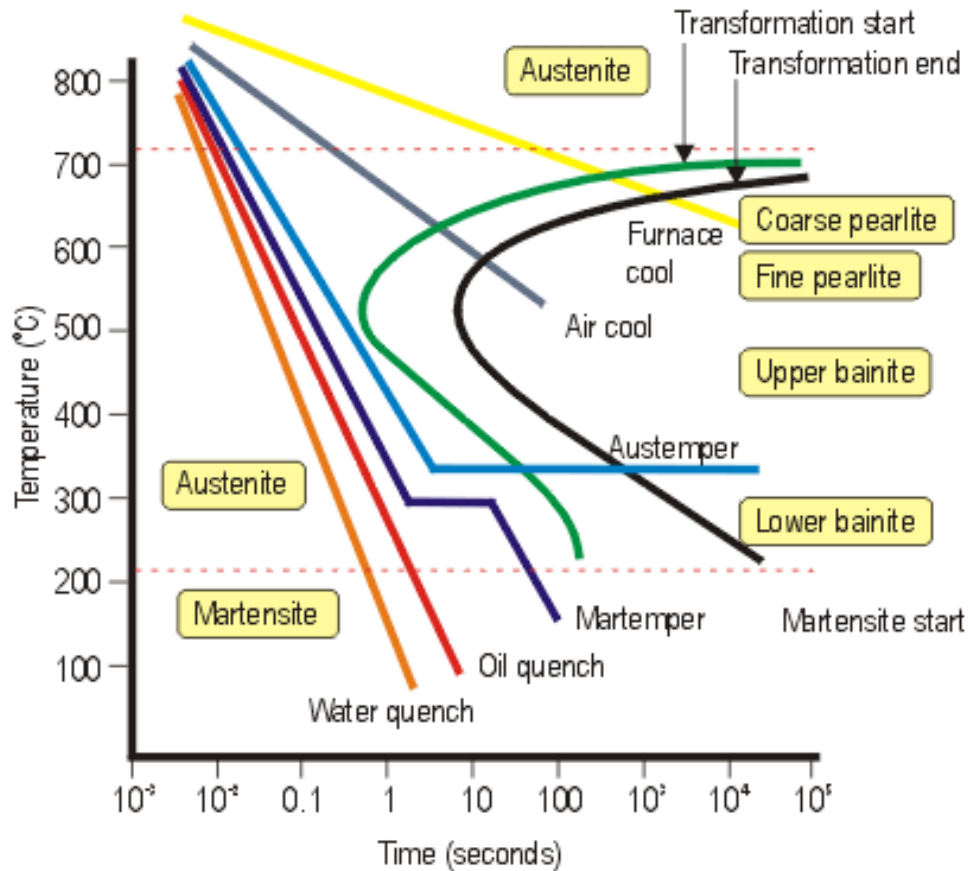


Figure 13: Continuous Cooling Transformation diagram showing transformation during continuous cooling.

5.5 CCT Diagram Superimposed on I.T. Diagram

When the CCT curve is compared with the I.T. / TTT diagram, the CCT start and finish curves are generally shifted to longer times and lower temperatures. This is because transformation during continuous cooling is different from transformation at a fixed isothermal holding temperature.

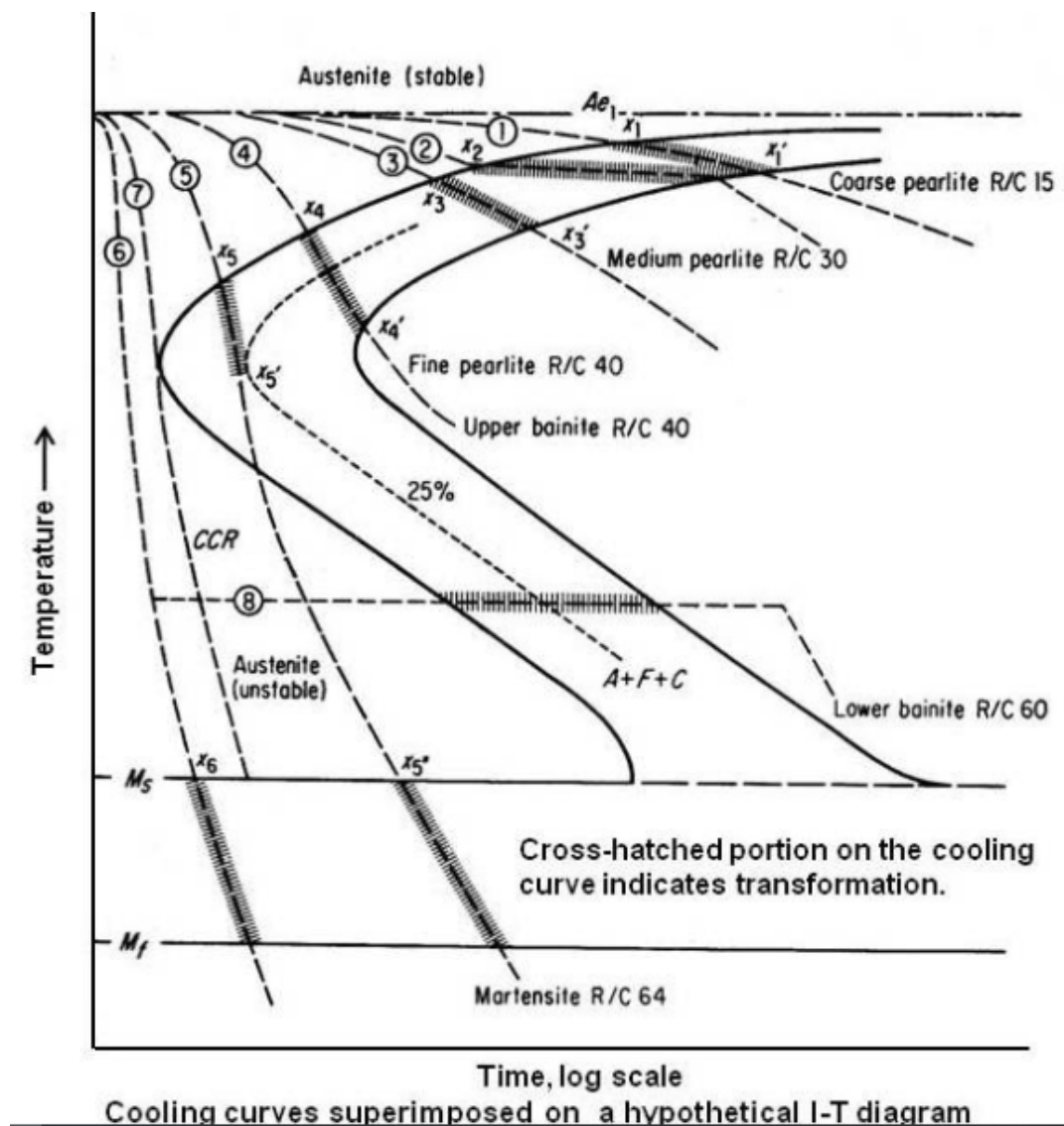


Figure 14: CCT diagram superimposed on I.T. / TTT diagram.

5.6 Difference Between I.T. and CCT Diagrams

Point	I.T. / TTT diagram	CCT diagram
Full form	Isothermal Transformation / Time-Temperature-Transformation diagram	Continuous Cooling Transformation diagram
Temperature condition	Constant temperature holding	Continuously decreasing temperature
Cooling condition	Rapid cooling to a selected temperature and holding	Continuous cooling from austenite region
Nature	More theoretical and basic	More practical for industrial heat treatment
Use	To understand transformation mechanism	To select actual cooling rate and heat treatment route

Linked processes	Austempering and martempering understanding	Annealing, normalising, hardening practice
------------------	---	--

Exam point: I.T. diagram: constant-temperature transformation. CCT diagram: transformation during continuous cooling.

6 Austempering and Martempering

6.1 Austempering

Austempering is an isothermal heat treatment process in which steel is heated to the austenite region, rapidly cooled to a temperature above M_s , held at that temperature until bainite forms and then cooled to room temperature.

Steps:

1. Heat steel to the austenitic region.
2. Hold until complete austenite formation.
3. Quench rapidly into a salt bath above M_s .
4. Hold isothermally until austenite transforms into bainite.
5. Cool to room temperature.

Final structure: bainite.

Advantages: good toughness, good strength, less distortion and less cracking compared with direct quenching.

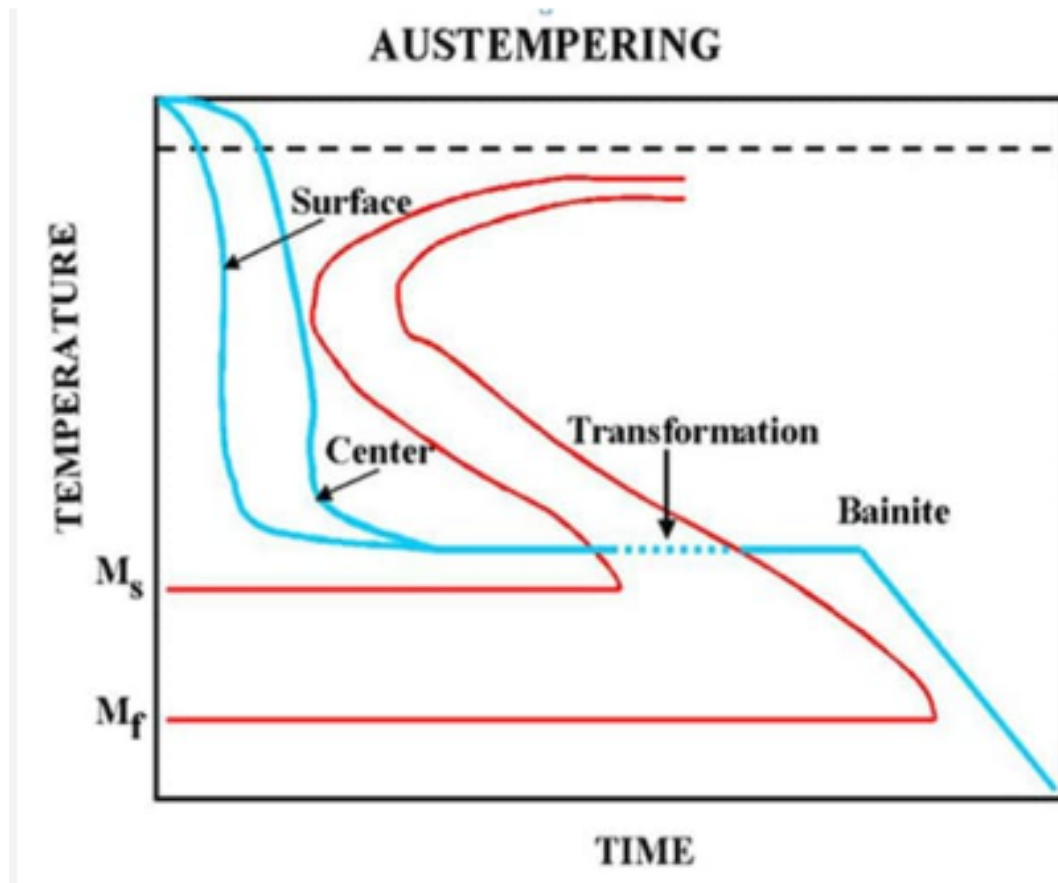


Figure 15: Austempering process path on I.T. / TTT diagram.

6.2 Martempering

Martempering is a heat treatment process in which steel is heated to the austenite region, rapidly cooled to a temperature just above M_s , held for temperature equalisation and then cooled through M_s and M_f to form martensite.

Steps:

1. Heat steel to the austenitic region.
2. Quench rapidly into a bath just above M_s .
3. Hold for surface and core temperature equalisation.
4. Cool through M_s and M_f to form martensite.
5. Tempering is usually done after martempering.

Final structure: martensite, usually tempered later.

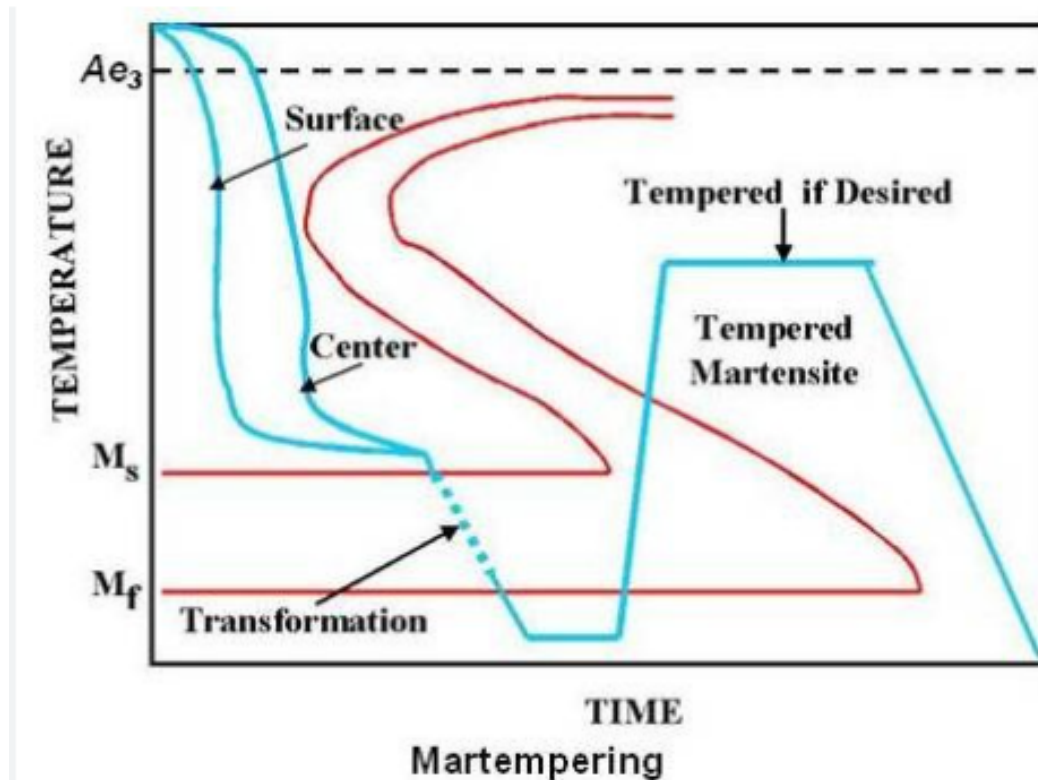


Figure 16: Martempering process path on I.T. / TTT diagram.

6.3 Comparison of Austempering and Martempering

Point	Austempering	Martempering
Holding temperature	Above M_s in bainite region	Just above M_s
Holding purpose	Complete bainite formation	Equalise temperature between surface and core
Final structure	Bainite	Martensite, usually tempered
Tempering need	Usually not necessary	Usually required
Main advantage	Strength with toughness and less distortion	High hardness with less distortion and cracking

7 Hardenability and Jominy End Quench Test

7.1 Hardenability

Hardenability is the ability of steel to develop hardness to a certain depth during quenching. It is also described as the ability of steel to form martensite throughout its cross-section during cooling.

Hardenability = depth of hardening.

7.2 Hardness and Hardenability

Point	Hardness	Hardenability
Meaning	Resistance to indentation, scratching or wear	Ability to harden through depth
Measured at	Surface or a point	From surface towards core
Depends mainly on	Carbon content and microstructure	Alloying elements, grain size, cooling rate and section size
Example	Rockwell hardness value	Depth up to which martensite forms

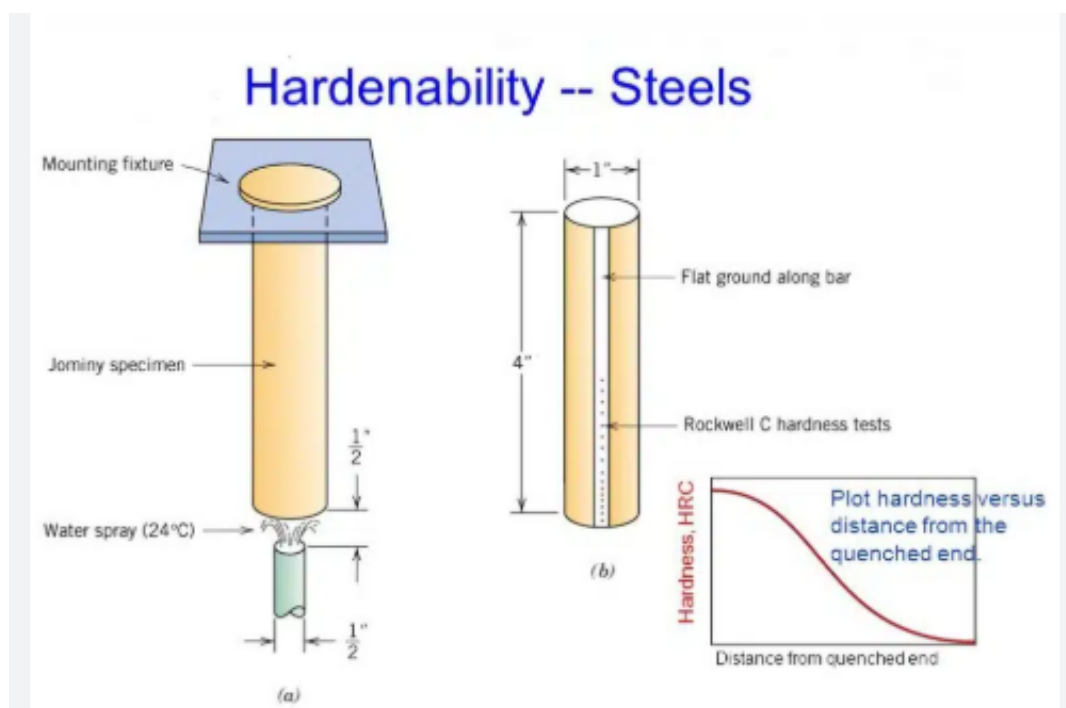


Figure 17: Low and high hardenability showing hardness distribution from surface to core.

7.3 Factors Affecting Hardenability

Hardenability depends on:

1. Carbon content.
2. Alloying elements such as Mn, Cr, Ni, Mo, V and W.
3. Austenite grain size.
4. Cooling rate.
5. Section thickness.
6. Quenching medium.

Alloying elements improve hardenability by delaying pearlite and bainite transformations. In transformation diagrams, this is usually represented by shifting the transformation curves to the right.

7.4 Jominy End Quench Test

The Jominy end quench test is a standard test used to determine the hardenability of steel by measuring hardness along the length of a quenched specimen.

Principle: One end of the heated steel specimen is quenched by a water jet. The quenched end cools fastest and the far end cools slowly. Hardness is measured at regular distances from the quenched end.

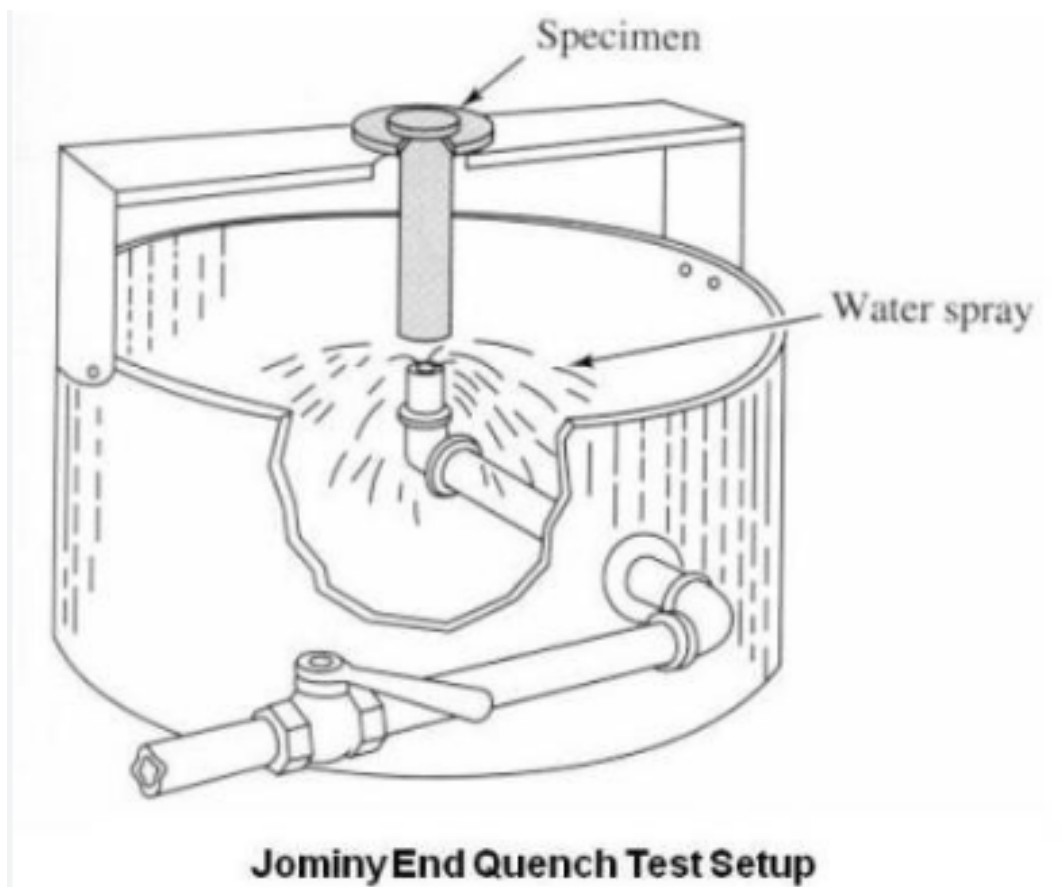


Figure 18: Jominy end quench test setup showing water jet at one end of the specimen.

7.5 Jominy Test Procedure

1. Prepare a standard cylindrical steel specimen.
2. Heat the specimen to the austenitising temperature.
3. Hold until complete austenite formation.
4. Place the specimen vertically in the Jominy fixture.
5. Spray water at one end only.

6. Cool the specimen.
7. Grind flat surfaces along the specimen length.
8. Measure Rockwell hardness at regular distances from the quenched end.
9. Plot hardness versus distance from the quenched end.

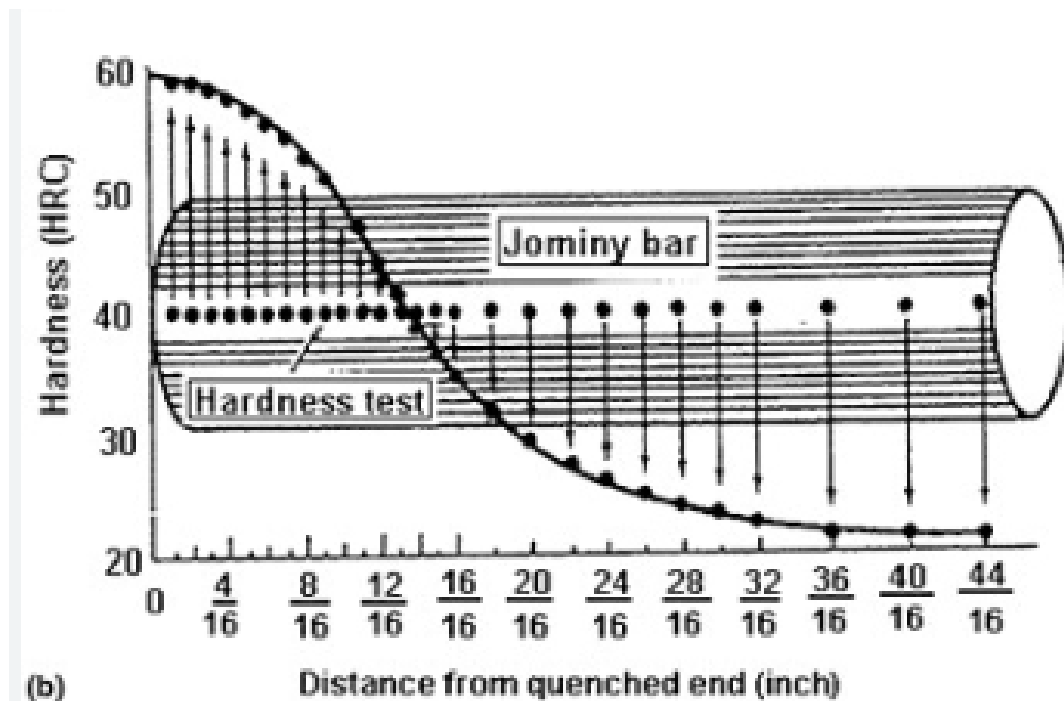


Figure 19: Typical Jominy hardness profile: hardness decreases with distance from quenched end.

7.6 Interpretation of Jominy Curve

- **High hardenability steel:** hardness remains high even at longer distances from the quenched end.
- **Low hardenability steel:** hardness drops rapidly with distance from the quenched end.

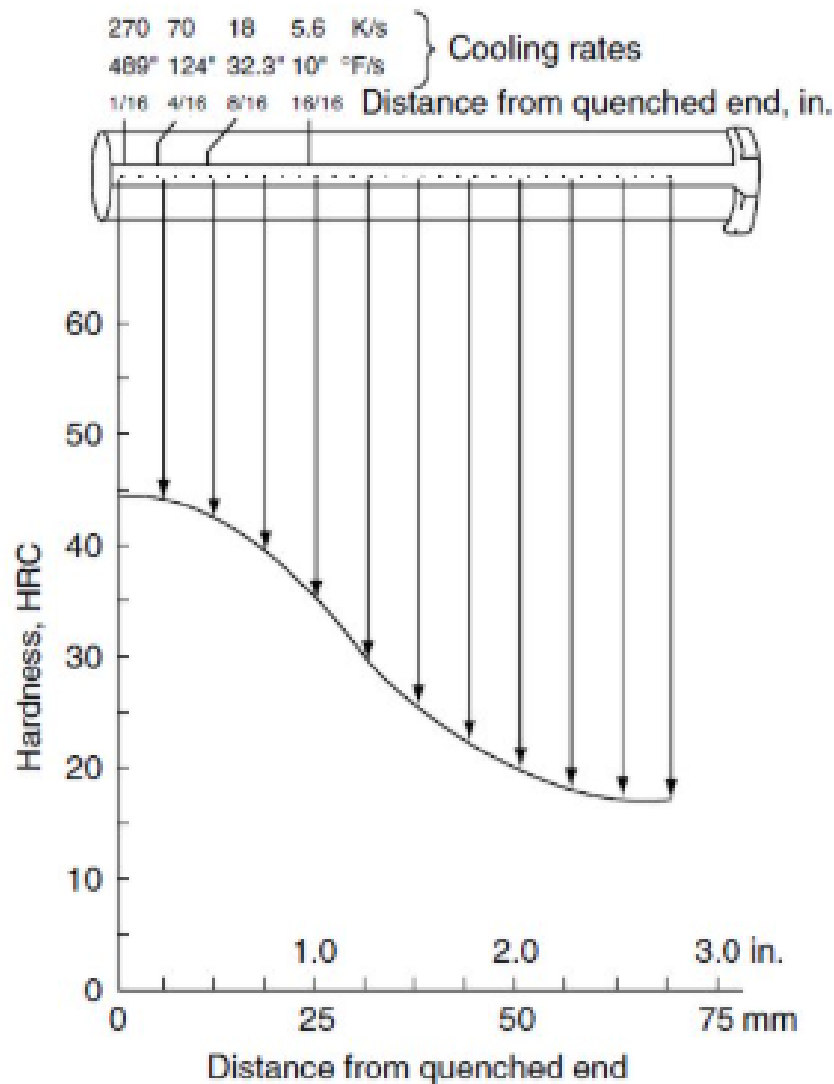


Figure 20: Comparison of Jominy curves for low and high hardenability steels.

8 Case Hardening and Diffusion Treatments

8.1 Case Hardening

Case hardening is a surface heat treatment process used to produce a hard outer surface and a tough inner core in steel components.

Need for case hardening:

- hard surface for wear resistance,
- tough core for shock resistance,
- improved fatigue resistance,
- longer service life.

Common applications: gears, cams, shafts, crankshafts, pins, bearings and machine parts.

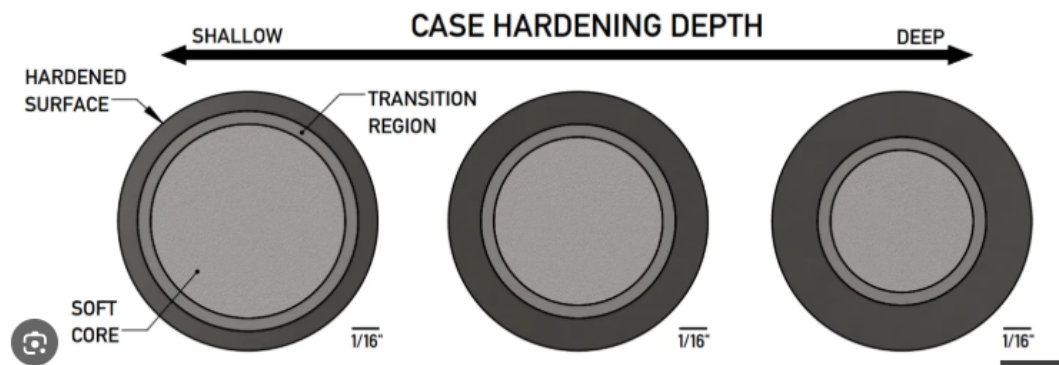


Figure 21: Case hardened component showing hard outer case and tough core.

8.2 Carburizing

Carburizing is a case hardening process in which carbon is added to the surface of low-carbon steel by heating it in a carbon-rich environment.

Process:

1. Low-carbon steel is heated in a carbon-rich medium.
2. Carbon diffuses into the surface.
3. The component is quenched.
4. The carbon-rich surface becomes hard martensite.
5. The low-carbon core remains tough.

Types: pack carburizing, gas carburizing and liquid carburizing.

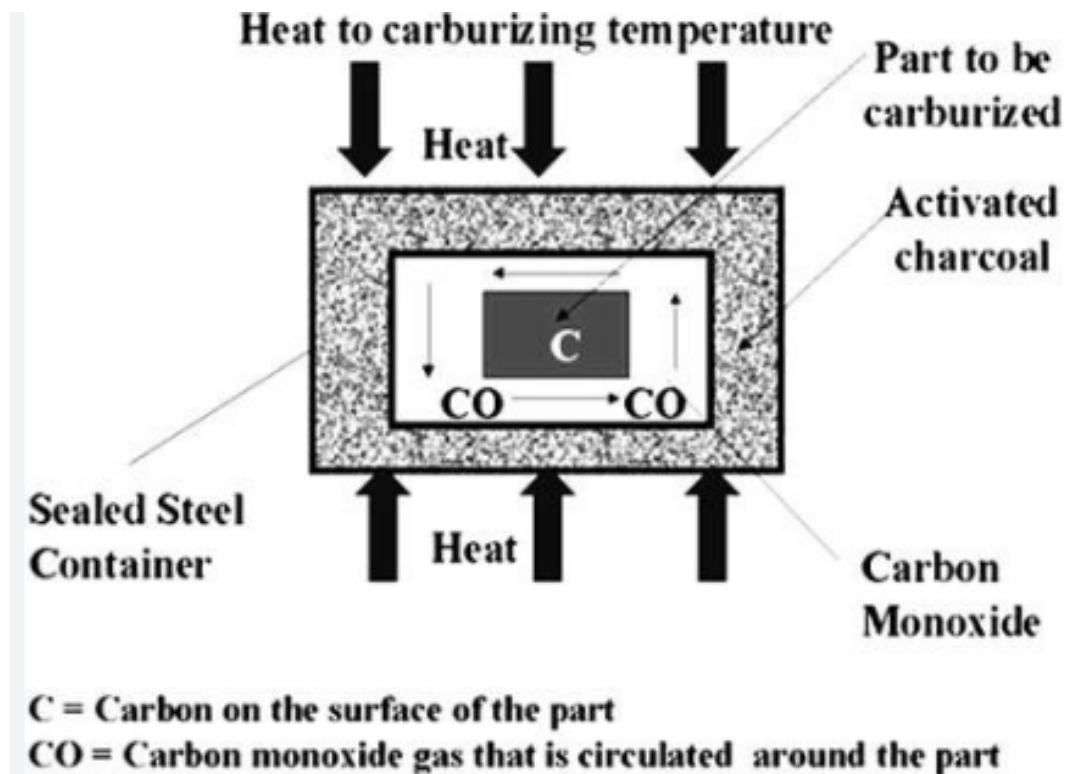


Figure 22: Carburizing process showing carbon diffusion into the surface of low-carbon steel.

8.3 Nitriding

Nitriding is a case hardening process in which nitrogen is introduced into the surface of steel to form hard nitrides.

Important points:

- Usually carried out in an ammonia or nitrogen-rich atmosphere.
- Produces very hard surface layer.
- Gives good wear resistance and fatigue resistance.
- Usually does not require quenching.
- Distortion is lower compared with carburizing because of lower processing temperature and absence of quenching.

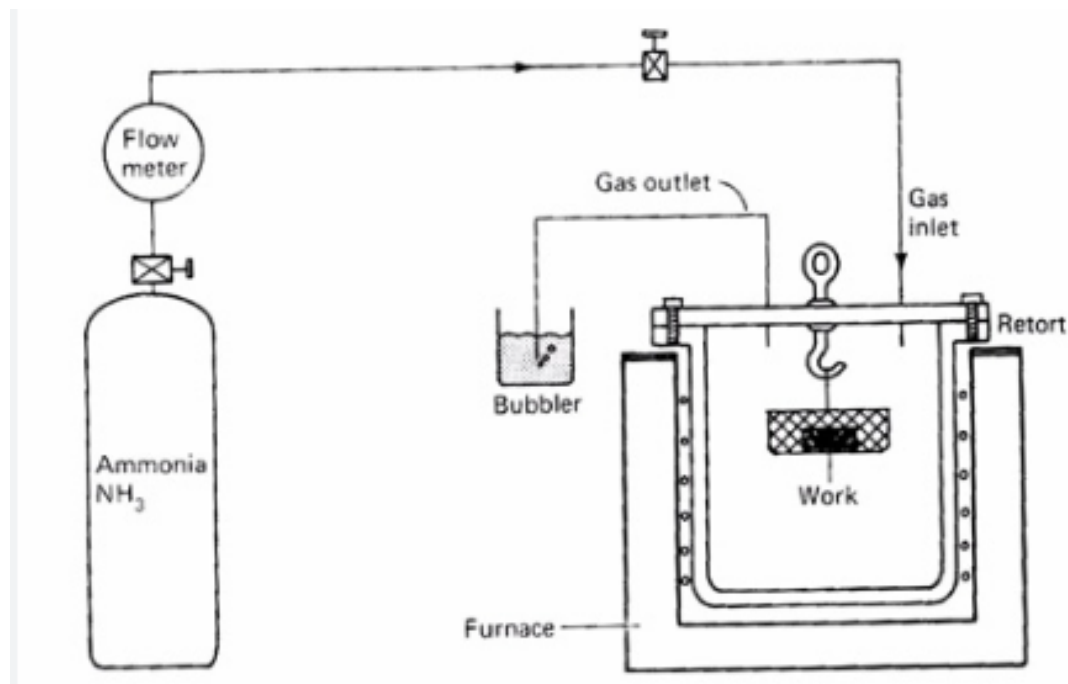


Figure 23: Nitriding process showing nitrogen diffusion and formation of hard nitride layer.

8.4 Cyaniding

Cyaniding is a case hardening process in which both carbon and nitrogen are introduced into the steel surface by heating in a cyanide salt bath.

Features:

- Adds carbon and nitrogen to the surface.
- Produces a thin and hard case.
- Faster than carburizing for small parts.
- Requires strict safety control because cyanide salts are highly toxic.

Applications: small gears, screws, bolts, pins and small machine parts.

8.5 Carbonitriding

Carbonitriding is a case hardening process in which both carbon and nitrogen are introduced into the steel surface, usually in a gaseous atmosphere.

Important points:

- Similar to carburizing but nitrogen is added along with carbon.
- Suitable for low-carbon steels.
- Produces a hard, wear-resistant surface.
- Improves fatigue strength.
- Commonly followed by quenching.

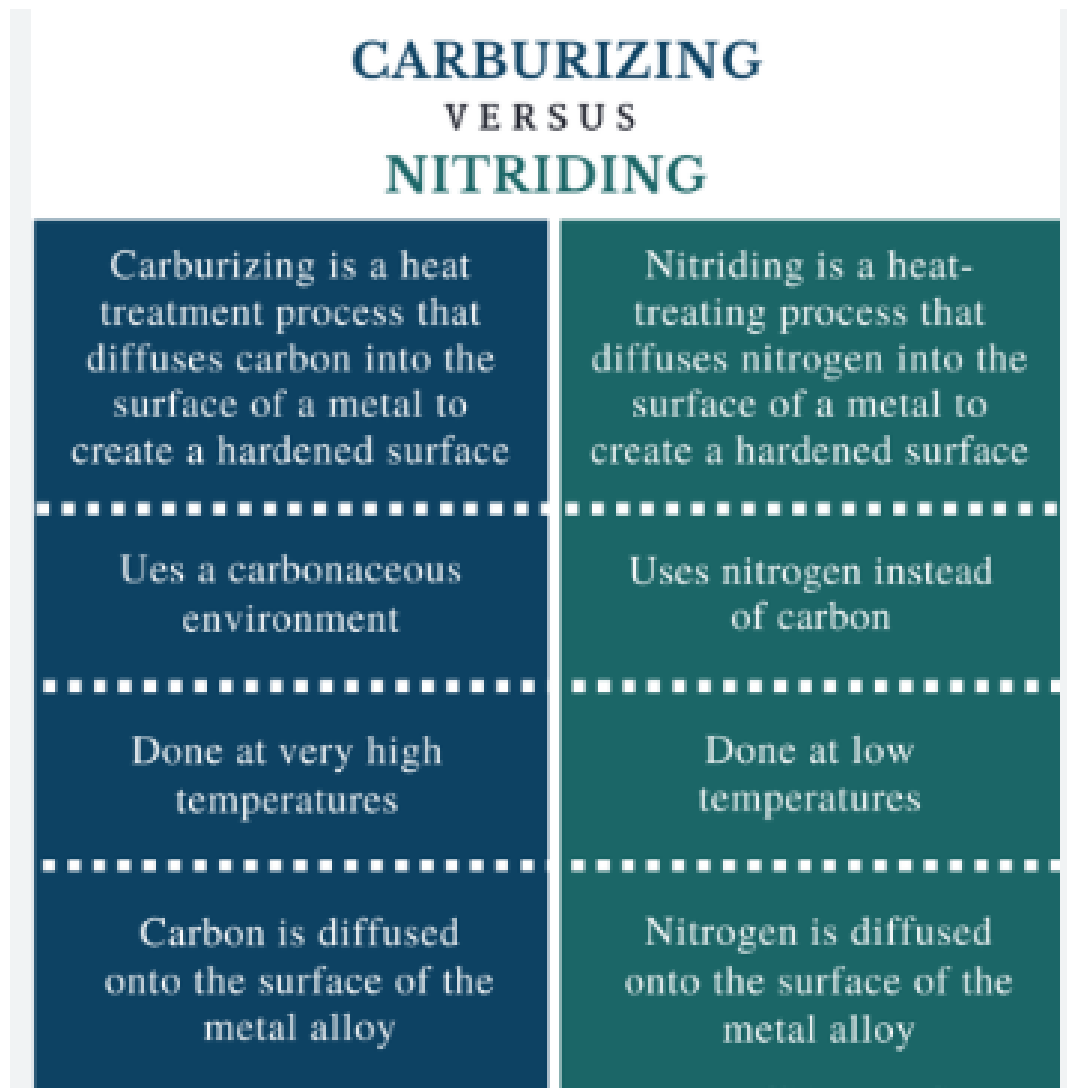


Figure 24: Comparison of carburizing, nitriding, cyaniding and carbonitriding based on diffusing elements.

8.6 Comparison of Case Hardening Methods

Process	Element added	Medium	Typical applications
Carburizing	Carbon	Solid, gas or liquid carbon medium	Gears, cams, shafts
Nitriding	Nitrogen	Ammonia or nitrogen atmosphere	Dies, gears, crankshafts, cylinder liners
Cyaniding	Carbon + nitrogen	Cyanide salt bath	Small gears, bolts, pins
Carbonitriding	Carbon + nitrogen	Gas atmosphere	Gears, shafts, fasteners, small precision parts

9 Flame, Induction, Vacuum and Plasma Hardening

9.1 Flame Hardening

Flame hardening is a surface hardening process in which the surface of steel is heated rapidly by an oxy-acetylene flame and then immediately quenched to produce a hard martensitic surface.

Steps:

1. Heat the surface using oxy-acetylene flame.
2. Raise the surface above critical temperature.
3. Quench immediately using water spray.
4. Martensite forms at the surface.
5. Core remains relatively tough.

Applications: large gears, shafts, rails, crane wheels, sprockets and machine ways.

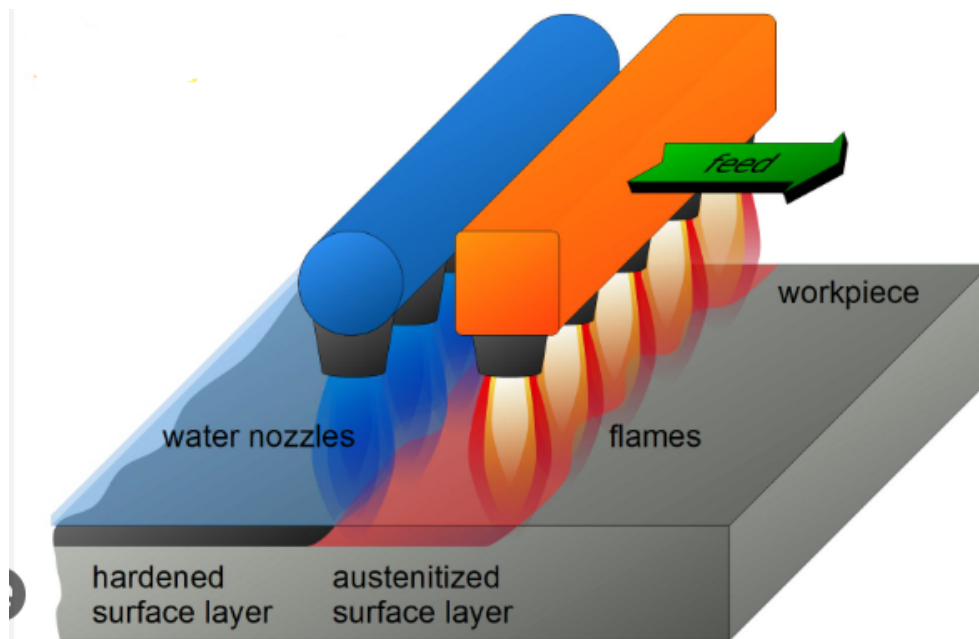


Figure 25: Flame hardening setup showing flame heating and water spray quenching.

9.2 Induction Hardening

Induction hardening is a surface hardening process in which the surface of steel is heated by electromagnetic induction and then rapidly quenched to obtain a hard martensitic surface.

Principle:

Alternating current → magnetic field → eddy currents → surface heating

Advantages: fast heating, good control of case depth, less oxidation, suitable for automation and high production rate.

Applications: gears, crankshafts, camshafts, axles, shafts, bearing surfaces and pins.

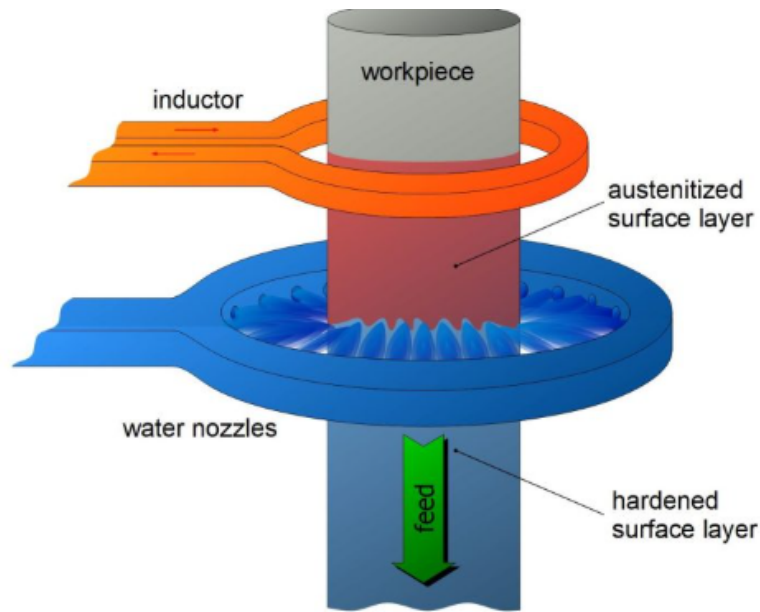


Figure 26: Induction hardening setup showing induction coil, component and quenching.

9.3 Difference Between Flame Hardening and Induction Hardening

Point	Flame hardening	Induction hardening
Heating source	Oxy-acetylene flame	Electromagnetic induction
Control	Moderate; depends on operator and torch movement	Better and more repeatable control
Speed	Slower than induction	Very fast
Equipment cost	Lower	Higher
Surface oxidation	More possibility	Less possibility
Suitable for	Large or irregular surfaces	Repetitive and precision components

9.4 Vacuum Hardening

Vacuum hardening is a hardening process carried out in a vacuum furnace to prevent oxidation, scaling and decarburisation during heating.

Advantages:

- No oxidation.
- No scaling.
- No decarburisation.
- Clean and bright surface finish.

- Good dimensional control.
- Suitable for precision components.

Applications: tool steels, dies, moulds, aerospace parts, medical tools and precision machine components.

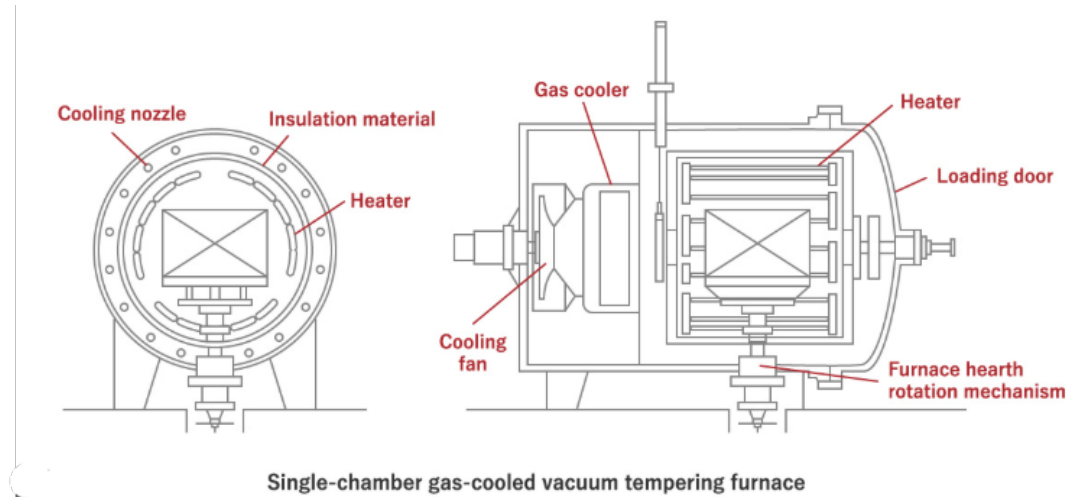


Figure 27: Vacuum hardening furnace showing heating in a controlled oxygen-free environment.

9.5 Plasma Hardening

Plasma hardening is a surface hardening process in which plasma energy or ionised gas is used to modify and harden the surface of a component. In many industrial cases, plasma nitriding or ion nitriding is used, where nitrogen ions form a hard nitride layer on the surface.

Steps in plasma nitriding:

1. Component is placed inside a vacuum chamber.
2. Low-pressure nitrogen-containing gas is introduced.
3. Electric potential is applied.
4. Plasma or glow discharge is produced.
5. Nitrogen ions react with the surface.
6. Hard nitride layer is formed.

Applications: gears, dies, tools, crankshafts, injection moulding parts, automotive and aerospace components.

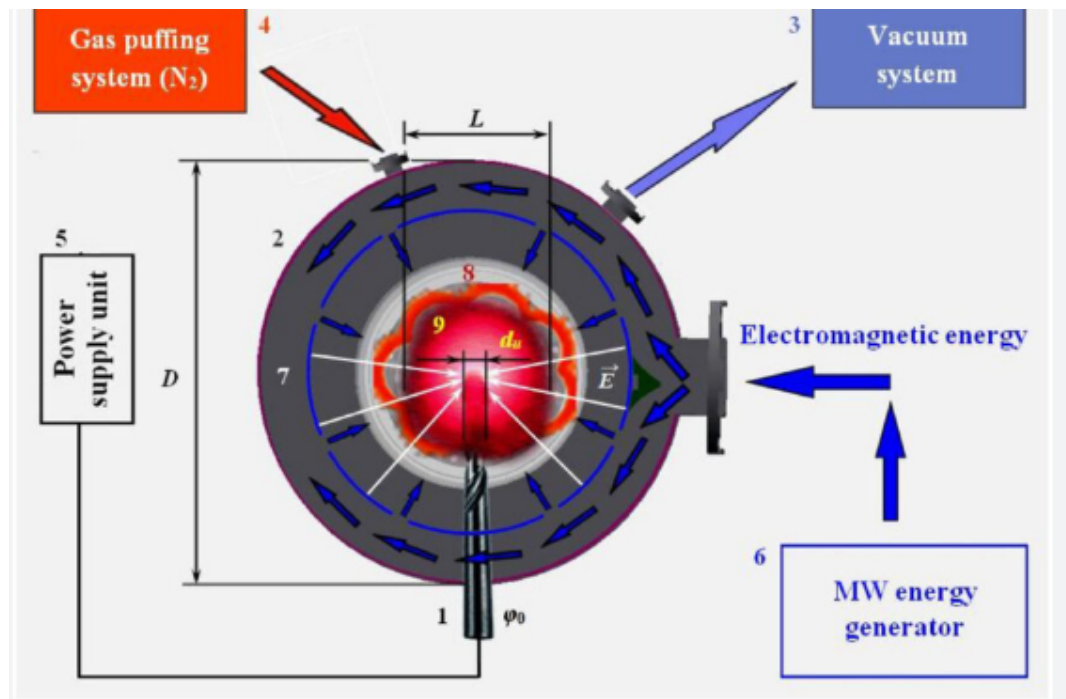


Figure 28: Plasma hardening chamber showing ionised gas and surface modification.

9.6 Vacuum Hardening and Plasma Hardening - Comparison

Point	Vacuum hardening	Plasma hardening
Main purpose	Hardening without oxidation	Surface hardening using ionised gas/plasma
Medium	Vacuum furnace	Low-pressure ionised gas
Surface change	Mainly clean hardening	Surface modification or nitride layer formation
Distortion	Low	Low
Applications	Tools, dies, precision parts	Wear and fatigue resistant surfaces

10 Thermo-Mechanical Treatment

Thermo-mechanical treatment is a process in which plastic deformation and heat treatment are combined to improve the mechanical properties and microstructure of metals.

Thermal treatment + Mechanical deformation = Thermo-mechanical treatment

Basic principle: TMT controls temperature, deformation, recrystallisation, grain growth and cooling rate to obtain fine grains and improved properties.

Steps:

1. Heat the metal to a suitable temperature.

2. Apply mechanical deformation such as rolling or forging.
3. Control recrystallisation and grain growth.
4. Apply controlled cooling.
5. Obtain refined microstructure and improved strength and toughness.

Applications: steel plates, rails, automotive sheets, pipelines, HSLA steels and aerospace alloys.

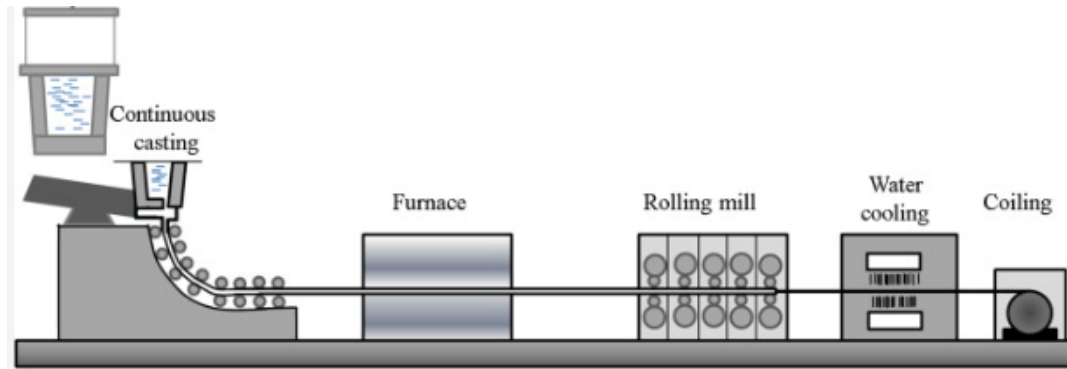


Figure 29: Thermo-mechanical treatment flowchart showing heating, deformation and controlled cooling.

11 Elementary Ideas on Sintering

Sintering is a process in which compacted metal or ceramic powder is heated below its melting point so that particles bond together and form a solid component.

Powder + Compaction + Heating below melting point → Solid component

11.1 Basic Steps in Powder Metallurgy and Sintering

1. Powder production.
2. Powder blending or mixing.
3. Compaction in die.
4. Sintering below melting point.
5. Finishing operation if required.

11.2 What Happens During Sintering?

During sintering:

- particles bond together,
- porosity decreases,
- strength increases,

- density increases,
- diffusion occurs between particles,
- shape is retained.

Applications: gears, bearings, filters, cutting tool inserts, magnets, ceramics and self-lubricating bearings.

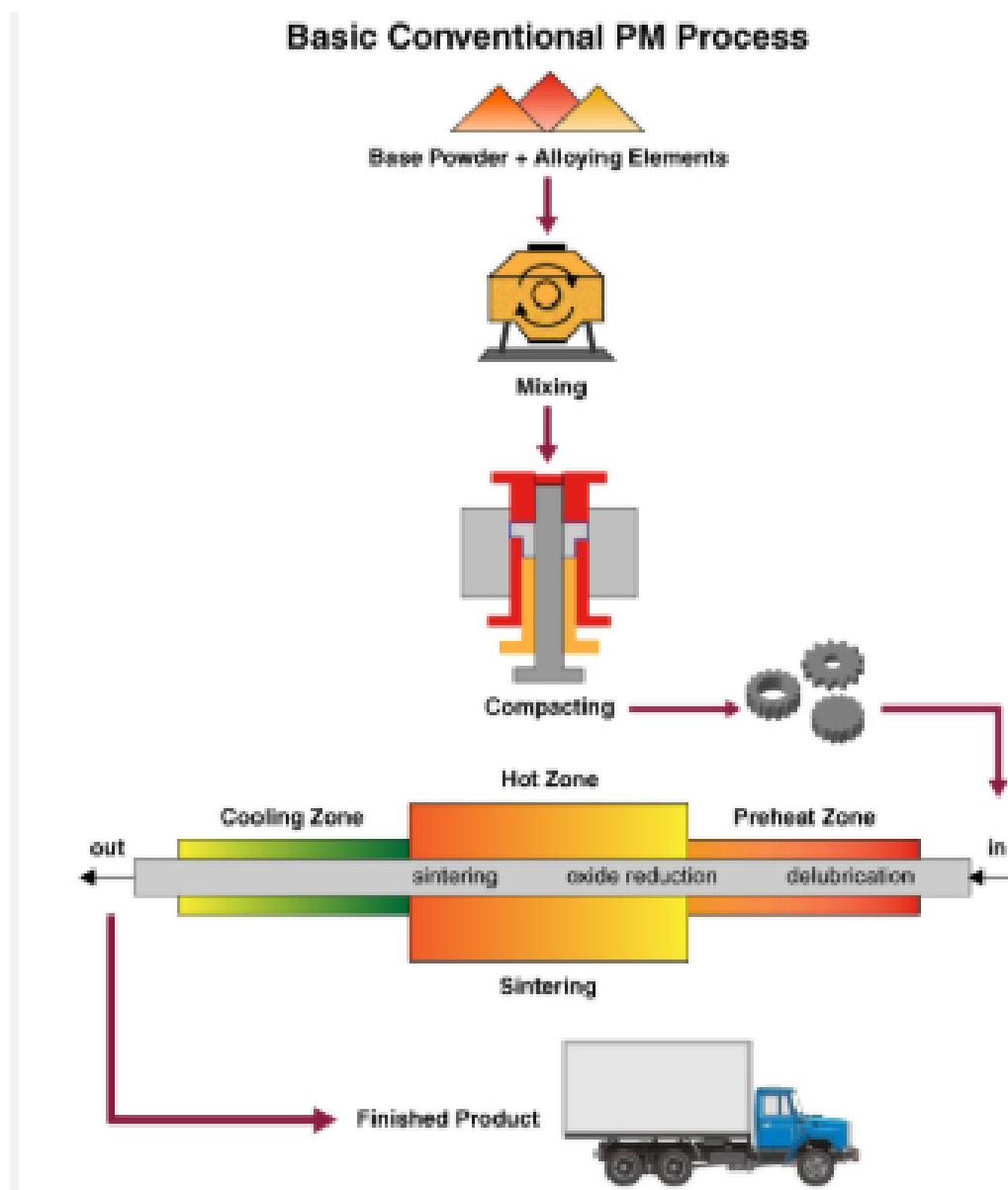


Figure 30: Sintering process showing loose powder, green compact and sintered component.

12 Important Exam-Oriented Points

12.1 Very Short Answer Points

1. Heat treatment is controlled heating and cooling in the solid state.
2. Annealing generally involves slow cooling and produces soft structures.

3. Full annealing improves softness, ductility and machinability.
4. Stress relief annealing is done below A_1 to remove residual stresses.
5. Recrystallisation annealing restores ductility in cold-worked metals.
6. Spheroidising improves machinability of high-carbon steels.
7. Normalising uses air cooling and refines grains.
8. Hardening produces martensite by rapid quenching.
9. Tempering reduces brittleness of hardened steel.
10. TTT diagram shows transformation at constant temperature.
11. CCT diagram shows transformation during continuous cooling.
12. Austempering produces bainite.
13. Martempering produces martensite with less distortion.
14. Hardenability means depth of hardening.
15. Jominy end quench test is used to determine hardenability.
16. Case hardening gives a hard surface and tough core.
17. Carburizing adds carbon to the surface.
18. Nitriding adds nitrogen to the surface.
19. Cyaniding and carbonitriding add both carbon and nitrogen.
20. Sintering bonds powder particles below the melting point.

12.2 Important Two-Mark Questions

1. Define heat treatment.
2. What are the three basic steps in heat treatment?
3. Define full annealing.
4. Define stress relief annealing.
5. What is recrystallisation annealing?
6. What is spheroidising?
7. Differentiate annealing and normalising.
8. Define hardening and tempering.
9. What is an I.T. / TTT diagram?
10. What is a CCT diagram?
11. Define austempering and martempering.
12. Define hardenability.
13. What is the purpose of Jominy end quench test?

14. Define case hardening.
15. Differentiate carburizing and nitriding.
16. Define flame hardening and induction hardening.
17. What is vacuum hardening?
18. What is plasma hardening?
19. Define thermo-mechanical treatment.
20. Define sintering.

12.3 Important Five-Mark Questions

1. Explain the basic principles and purpose of heat treatment.
2. Explain full annealing and stress relief annealing.
3. Explain recrystallisation and spheroidising annealing.
4. Compare annealing, normalising, hardening and tempering.
5. Explain I.T. diagram and its important regions.
6. Explain cooling curves superimposed on I.T. diagram.
7. Differentiate I.T. diagram and CCT diagram.
8. Explain austempering and martempering.
9. Explain hardenability and Jominy end quench test.
10. Explain carburizing, nitriding, cyaniding and carbonitriding.
11. Compare flame hardening and induction hardening.
12. Explain vacuum hardening and plasma hardening.
13. Explain thermo-mechanical treatment.
14. Explain sintering with steps and applications.

12.4 Important Long-Answer Questions

1. Explain different annealing processes: full annealing, stress relief annealing, recrystallisation annealing and spheroidising.
2. Explain normalising, hardening and tempering of steel with neat heat treatment cycles.
3. Explain I.T. / TTT diagram, cooling curves superimposed on I.T. diagram and CCT diagram.
4. Compare austempering and martempering based on cooling path, final structure, advantages and applications.
5. Explain hardenability and the Jominy end quench test with neat diagram and hardness curve.

6. Explain case hardening processes: carburizing, nitriding, cyaniding and carbonitriding.
7. Explain flame hardening, induction hardening, vacuum hardening and plasma hardening with applications.
8. Explain thermo-mechanical treatment and elementary ideas on sintering.

13 Unit Summary

- Unit II deals with microstructure control by heat treatment.
- Heating temperature, soaking time and cooling rate decide final properties.
- Annealing softens steel, while normalising refines grains and improves strength.
- Hardening produces martensite, and tempering reduces brittleness.
- I.T. and CCT diagrams help predict transformation products during cooling.
- Austempering produces bainite; martempering produces martensite with reduced distortion.
- Hardenability describes the depth to which steel can be hardened.
- Jominy end quench test is used to compare hardenability of steels.
- Case hardening produces a hard surface and tough core.
- Flame and induction hardening are selective surface hardening processes.
- Vacuum and plasma hardening are modern controlled hardening methods.
- Thermo-mechanical treatment combines deformation and heat treatment.
- Sintering bonds compacted powder particles below the melting point.

14 Glossary of Important Terminologies

The following short definitions are useful for quick revision and two-mark examination answers.

Term	Short definition / exam point
Heat treatment	Controlled heating and cooling of a metal or alloy in solid state to change microstructure and properties.
Soaking	Holding a component at a selected temperature for uniform heating and transformation.
Annealing	Heating, holding and slow cooling to soften steel and improve ductility or machinability.
Full annealing	Heating steel above critical temperature and furnace cooling to obtain a soft structure.
Stress relief annealing	Heating below A_1 to remove residual stresses without major microstructural change.

Term	Short definition / exam point
Recrystallisation	Formation of new strain-free grains in a cold-worked metal during heating.
Spheroidising	Annealing process that converts lamellar cementite into spherical cementite particles.
Normalising	Heating above critical temperature and cooling in still air to refine grains.
Hardening	Austenitising followed by rapid quenching to produce martensite.
Tempering	Reheating hardened steel below A_1 to reduce brittleness and improve toughness.
Austenite	FCC solid solution of carbon in gamma iron, stable at elevated temperatures.
Pearlite	Lamellar mixture of ferrite and cementite.
Bainite	Fine mixture of ferrite and cementite formed at intermediate transformation temperatures.
Martensite	Very hard and brittle phase formed by rapid quenching of austenite.
TTT diagram	Time-Temperature-Transformation diagram showing isothermal transformation of austenite.
CCT diagram	Continuous Cooling Transformation diagram showing austenite transformation during continuous cooling.
Critical cooling rate	Minimum cooling rate required to avoid the nose of TTT diagram and form martensite.
Austempering	Isothermal treatment that transforms austenite into bainite.
Martempering	Interrupted quenching process that forms martensite with reduced distortion.
Hardenability	Ability of steel to develop hardness through depth during quenching.
Jominy test	Standard end-quench test used to determine hardenability of steel.
Case hardening	Process that produces hard outer case and tough inner core.
Carburizing	Addition of carbon to the surface of low-carbon steel.
Nitriding	Addition of nitrogen to steel surface to form hard nitrides.
Cyaniding	Addition of carbon and nitrogen using a cyanide salt bath.
Carbonitriding	Addition of carbon and nitrogen, usually using a gaseous atmosphere.
Flame hardening	Surface hardening by flame heating followed by quenching.
Induction hardening	Surface hardening by electromagnetic induction heating followed by quenching.
Vacuum hardening	Hardening in a vacuum furnace to prevent oxidation and decarburisation.
Plasma hardening	Surface hardening using ionised gas or plasma.

Term	Short definition / exam point
Thermo-mechanical treatment	Combined deformation and heat treatment to improve microstructure and properties.
Sintering	Heating compacted powder below melting point to bond particles and form a solid component.
